

# 1.2 Atoms, Molecules & Stoichiometry

## Question Paper

|            |                                      |
|------------|--------------------------------------|
| Course     | CIEA Level Chemistry                 |
| Section    | 1. Physical Chemistry                |
| Topic      | 1.2 Atoms, Molecules & Stoichiometry |
| Difficulty | Hard                                 |

**Time allowed:** 70

**Score:** /57

**Percentage:** /100

### Question 1a

Magnesium carbonate is a common antacid that is used to react with and neutralise excess stomach acid, HCl.



The total mass of the reactants is 9.91 g.

Calculate the volume of carbon dioxide that is produced at room temperature and pressure.

Volume of  $\text{CO}_2$  = .....  $\text{dm}^3$

[4 marks]

### Question 1b

Another magnesium salt, magnesium sulfate, can exist in hydrated or anhydrous forms.

6.31 g of a sample of magnesium sulphate,  $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ , was strongly heated until a mass of 3.61 g of solid remained with no further change in mass recorded.

Calculate the number of moles of water of crystallisation,  $x$ , in the magnesium sulphate sample.

moles of water = ..... mol

[3 marks]

**Question 1c**

Another sample of hydrated magnesium sulphate,  $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ , is found to contain 51.1% water by mass.

Calculate the number of moles of water of crystallisation,  $x$ , in this sample.

moles of water = ..... mol

**[3 marks]**

**Question 1d**

Describe the effect on the value of  $x$  obtained if the salt was not heated strongly to a constant mass.

**[2 marks]**

**Question 2a**

Fireworks contain elements to give them their colour, as well as oxygen-containing compounds to facilitate speedy combustion.

In a firework, solid potassium nitrate,  $\text{KNO}_3$ , decomposes to form solid potassium nitrite,  $\text{KNO}_2$ , and oxygen,  $\text{O}_2$ .

Calculate the mass, in g, of potassium nitrate,  $\text{KNO}_3$ , required to make 1.5 g of oxygen. Give your answer to 2 significant figures.

mass of  $\text{KNO}_3$  = ..... g

**[4 marks]**

### Question 2b

Calculate the volume, in  $\text{dm}^3$ , of gas that is produced when solid potassium nitrate,  $\text{KNO}_3$ , decomposes at room temperature and pressure. Give your answer to 2 significant figures.

volume of gas = .....  $\text{dm}^3$

[1 mark]

### Question 2c

Potassium can form a superoxide,  $\text{KO}_2$  (s), which will react with carbon dioxide,  $\text{CO}_2$  (g), to produce potassium carbonate,  $\text{K}_2\text{CO}_3$  (s) and oxygen,  $\text{O}_2$  (g).

Calculate the volume, in  $\text{dm}^3$ , of carbon dioxide which will react with 5.00 g of the superoxide. Give your answer to 3 significant figures.

volume of  $\text{CO}_2$  = .....  $\text{dm}^3$

[4 marks]

### Question 2d

2.61 g of potassium carbonate,  $\text{K}_2\text{CO}_3$ , was produced during the reaction in part (c).

Calculate the percentage yield of potassium carbonate. Give your answer to 2 decimal places.

% yield = ..... %

[3 marks]

**Question 3a**

The coca-cola company states that a 12 oz can of coke contains 34 mg of caffeine,  $M_r = 194.19 \text{ g mol}^{-1}$ .

Calculate the number of moles of caffeine in a can of coke.

moles of caffeine = ..... mol

[1 mark]

**Question 3b**

Caffeine is made up of 49.48% carbon, 5.20% hydrogen, 28.85% nitrogen and the rest is oxygen.

Calculate the empirical formula of caffeine.

[3 marks]

**Question 3c**

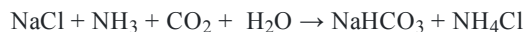
Calculate the molecular formula of caffeine using your answer from part (b).

[1 mark]

### Question 4a

This question is about the reactions of ionic compounds.

Sodium carbonate is manufactured in a two-stage process as shown.



Calculate the maximum mass of sodium carbonate that can be obtained from 0.75 kg of sodium chloride.

mass of  $\text{Na}_2\text{CO}_3$  = ..... (g)

**[3 marks]**

### Question 4b

Norgessalpeter,  $\text{Ca}(\text{NO}_3)_2$ , was the first nitrogen fertiliser to be manufactured in Norway. It is produced by the reaction of calcium carbonate with nitric acid.

In an experiment, an excess of powdered calcium carbonate was added to  $37.4 \text{ cm}^3$  of  $0.531 \text{ mol dm}^{-3}$  nitric acid.

Calculate the minimum mass of  $\text{CaCO}_3$  that should be added to react with all of the nitric acid. Give your answer to an appropriate number of significant figures.

mass of  $\text{CaCO}_3$  = .....

**[4 marks]**

**Question 4c**

A 0.0830 mol sample of pure zinc oxide was added to 75.0 cm<sup>3</sup> of 0.90 mol dm<sup>-3</sup> hydrochloric acid.

Assuming the reaction has a percentage yield of 60.6%, calculate the mass of anhydrous zinc chloride that could be obtained from the products of this reaction.

mass of zinc chloride = ..... g

**[4 marks]**

**Question 4d**

Magnesium nitride, Mg<sub>3</sub>N<sub>2</sub>, reacts with water to form magnesium hydroxide and ammonia.

Calculate the number of molecules of ammonia gas produced by the reaction of 12.62 g of magnesium nitride.

number of molecules = .....

**[5 marks]**

### Question 5a

Apatite is a general term used for a family of various calcium phosphate compounds commonly found in rocks and minerals, bones and teeth.

i)

State the formula of calcium phosphate.

[1]

ii)

Hydroxylapatite,  $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ , is one member of the apatite family and is a major component of bone.

Give the oxidation number of calcium in hydroxyapatite. Explain your answer.

[3]

iii)

Another member of the apatite family is chlorapatite.

Suggest the formula of chlorapatite.

[1]

**[5 marks]**



**Question 5b**

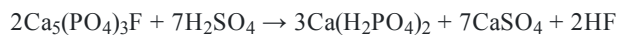
Fluorapatite,  $\text{Ca}_5(\text{PO}_4)_3\text{F}$ , reacts with sulfuric acid to form hydrogen phosphate, calcium sulfate and hydrogen fluoride.

Write the balanced chemical equation for this reaction.

[2 marks]

**Question 5c**

Fluorapatite,  $\text{Ca}_5(\text{PO}_4)_3\text{F}$ , can also react with sulfuric acid to form calcium dihydrogenphosphate,  $\text{Ca}(\text{H}_2\text{PO}_4)_2$ , calcium sulfate and hydrogen fluoride.



Calculate the mass of calcium dihydrogenphosphate formed when 50.2 g of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  reacts with 87.9 g  $\text{H}_2\text{SO}_4$ .

mass of  $\text{Ca}(\text{H}_2\text{PO}_4)_2 = \dots\dots\dots$  g

[5 marks]

## Model Answers: Hard

1a

The volume of CO<sub>2</sub> produced is:

- Mass ratio of 84.3 MgCO<sub>3</sub> : 73 HCl; [1 mark]
- Total relative mass = 157.3 so mass of MgCO<sub>3</sub> =  $\frac{9.91 \times 84.3}{157.3} = 5.31 \text{ g MgCO}_3$

**OR**

Total relative mass = 157.3 so mass of HCl =  $\frac{9.91 \times (36.5 \times 2)}{157.3} = 4.60 \text{ g} / 4.599 \text{ g HCl}$ ; [1 mark]

- Moles MgCO<sub>3</sub> =  $\frac{\text{mass}}{\text{molar mass}} = \frac{5.31}{84.3} = 0.063 \text{ (moles)}$

**OR**

Moles HCl =  $\frac{\text{mass}}{\text{molar mass}} = \frac{4.60}{36.5} = 0.126 \text{ (moles)}$ ; [1 mark]

- Volume CO<sub>2</sub> = moles CO<sub>2</sub> x molar gas volume = 0.063 x 24.0 = 1.51 dm<sup>3</sup>; [1 mark]

**[Total: 4 marks]**

- This question is based on scaling the total mass according to the ratio of relative masses of the reactants
- You need to know the mass of the individual reactants to be able to complete this question
- You can use the chemical equation to determine a ratio or percentage of magnesium carbonate to hydrochloric acid
- Step 1 - Use the chemical equation to calculate the mass of each reactant involved
  - MgCO<sub>3</sub> = 24.3 + 12.0 + (16.0 x 3) = 84.3
  - 2HCl = (2 x (1.0 + 35.5)) = 73
  - Total mass of reactants = 84.3 + 73 = 157.3
- Step 2 - Use the masses to determine the actual amount of each reactant
  - The second point on the mark scheme shows how to do this as a ratio
  - You can calculate the percentage of each reactant and apply that to the 9.91 g

- Step 3 - Calculate the moles of magnesium carbonate and hydrochloric acid
  - This is to determine the limiting reagent / reactant
  - The volume of carbon dioxide produced is dependent on the number of moles of limiting reagent / reactant
- Step 4 - Use the moles of limiting reagent / reactant to calculate the volume of carbon dioxide produced
  - **Remember:** One mole of any gas occupies  $24.0 \text{ dm}^3$  at room temperature

1b

To calculate the number of moles of water of crystallisation, x, in the magnesium sulphate sample:

- Moles of  $\text{MgSO}_4 = \frac{\text{mass}}{M_r} = \frac{3.61}{120.4} = 0.03 \text{ (moles)}; [1 \text{ mark}]$
- Moles  $\text{H}_2\text{O} = \frac{\text{mass}}{M_r} = \frac{6.31 - 3.61}{18.0} = 0.15 \text{ (moles)}; [1 \text{ mark}]$
- Ratio  $\text{MgSO}_4 : \text{H}_2\text{O} = 0.03 : 0.15 = 1 : 5$

**AND**

$$x = 5 / \text{formula} = \text{MgSO}_4 \cdot 5\text{H}_2\text{O}; [1 \text{ mark}]$$

**[Total: 3 marks]**

- When heated, the water of crystallisation evaporated as follows:
  - $\text{MgSO}_4 \cdot x\text{H}_2\text{O} \rightarrow \text{MgSO}_4 + x\text{H}_2\text{O}$
- This means that the question is essentially an empirical formula one where you need to work out the moles of the anhydrous  $\text{MgSO}_4$  and the moles of water, which come from the loss in mass during the experiment
- The key to accuracy in this experiment is that the hydrated salt has to be heated strongly to a constant mass, where all of the water has been driven off

1c

To calculate the number of moles of water of crystallisation, x, in this sample:

- Moles of  $\text{MgSO}_4 = \frac{\text{mass}}{M_r} = \frac{48.9}{120.4} = 0.4061 \text{ (moles)}; [1 \text{ mark}]$
- Moles  $\text{H}_2\text{O} = \frac{\text{mass}}{M_r} = \frac{51.1}{18.0} = 2.839 \text{ (moles)}; [1 \text{ mark}]$

- Ratio of  $\text{MgSO}_4 : \text{H}_2\text{O} = 0.4061 : 2.839 (= 6.99 : 1.00) = 7 : 1$

**AND**

$x = 7$  / formula =  $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ ; [1 mark]

**[Total: 3 marks]**

- Percentage composition values can be used in the same way as masses, where we assume that the sample mass is 100g and therefore 51.1% is equivalent to 51.1g

1d

The effect of not heating the hydrated salt strongly to constant mass is that:

- Some water remains with the salt/ the salt remains partially hydrated/ the salt is not fully anhydrous/ the mass of water lost is lower than it should be; [1 mark]
- The moles of water of crystallisation calculated are too low

**AND**

The value of  $x$  would be too small; [1 mark]

**[Total: 2 marks]**

- You need to think through the consequences of the calculations to do this question
  - If less mass appears to be lost that means fewer moles of water will be classed as the water of crystallisation
  - At the same time, a greater mass of the supposedly anhydrous salt will lead to a greater number of moles being calculated than are actually present
  - This means that the ratio will be more 'salt heavy' than it should be and fewer moles of water are calculated than are actually present

2a

The amount, in g, of potassium nitrate,  $\text{KNO}_3$ , required to make 1.5 g of oxygen,  $\text{O}_2$ , is:

- $2\text{KNO}_3 (\text{s}) \rightarrow 2\text{KNO}_2 (\text{s}) + \text{O}_2 (\text{g})$ ; [1 mark]
- Moles of oxygen =  $\frac{\text{mass}}{M_r} = \frac{1.5}{32.0} = 0.047 \text{ (mol)}$ ; [1 mark]
- Moles of  $\text{KNO}_3 = 0.047 \times 2 = 0.094 \text{ (mol)}$ ; [1 mark]
- Mass of  $\text{KNO}_3 = M_r \times \text{moles} = 101.1 \times 0.094 = 9.5 \text{ g (to 2 s.f.)}$ ; [1 mark]

**[Total: 4 marks]**

- When carrying out reacting mass calculations there are a few points to remember:
  - $M_r$  values should be given to 1 decimal place in line with the Periodic Table
  - Retain the numbers in your calculator throughout the working, otherwise your answer will not be close to the true value as small rounding errors can stack up quickly with moles and mass calculations
  - Check the stoichiometry of the equation that has been given, or you have written as this will influence the number of moles used in the calculation

2b

The volume of gas produced in the reaction in part (a) is:

- Volume of gas /  $\text{O}_2 = \text{mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 0.047 \times 24.0 = 1.1 \text{ dm}^3$  to 2 s.f.; **[1 mark]**

**[Total: 1 mark]**

- For this question, you are required to use the number of moles of gas which has been produced in the first reaction
- Therefore, to make sure you use the correct number of moles, it is a good idea to do the calculation again, to ensure you use the most accurate number of moles (e.g.  $1.5 \div 32.00 = 0.046875$ )
- You can then add this into your equation to calculate the volume ( $0.046875 \times 24.0 = 1.1$  to 2 decimal places) and then check what the required significant figures are, in this case, 2 s.f. means that the answer is  $1.1 \text{ dm}^3$
- Remember to calculate the volume of a gas, at RTP which you can assume in this question as there is not any extra information given such as pressure or temperature, you need to multiply the number of moles by  $24.0 \text{ dm}^3 \text{ mol}^{-1}$

2c

The volume of  $\text{CO}_2$  can be calculated by:

- $4\text{KO}_2 + 2\text{CO}_2 \rightarrow 2\text{K}_2\text{CO}_3 + 3\text{O}_2$ ; **[1 mark]**
- Moles  $\text{KO}_2$  in 5.00 g =  $\frac{\text{mass}}{M_r} = \frac{5.00}{71.1} = 0.0703 \text{ (mol) (3 s.f.)}$ ; **[1 mark]**
- Moles  $\text{CO}_2 = \frac{\text{moles of KO}_2}{2} = \frac{0.0703}{2} = 0.0352 \text{ (mol) (3 s.f.)}$ ; **[1 mark]**

- Volume of  $\text{CO}_2 = \text{mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 0.0352 \times 24.0 = 0.844 \text{ dm}^3$  to 3 s.f.; [1 mark]

**[Total: 4 marks]**

- **Careful:** Here you have been asked to give your answer to 3 significant figures, not 2 decimal places
- You should know that one mole of gas occupies  $24.0 \text{ dm}^3$  at room temperature and pressure
  - Since you are not given this information, you have to assume that the reaction is being performed at room temperature and pressure
- The steps are all relatively standard, but the question is less structured
- This means that you need to decide which steps to take and in which order
  - Always start with a balanced equation
  - Calculate what you know from the question
    - In this case, you know you start with 5.00 g of potassium superoxide which means that you can calculate the number of moles
  - Use the balanced equation to calculate what you need
    - In this case, you need to know the number of moles of  $\text{CO}_2$  to be able to calculate the volume of  $\text{CO}_2$

2d

To calculate the percentage yield:

- Moles  $\text{K}_2\text{CO}_3 = \frac{\text{Moles KO}_2}{2} = \frac{0.0703235}{2} = 0.0352 \text{ moles}$ ; [1 mark]
- Theoretical mass of  $\text{K}_2\text{CO}_3 = \text{mols} \times \text{molar mass} = 0.0352 \times 138.2 = 4.86 \text{ g} / 4.8593\dots \text{g}$ ; [1 mark]
- Percentage yield =  $\frac{\text{actual mass}}{\text{theoretical mass}} \times 100 = \frac{2.61}{4.86} \times 100 = 53.71 \%$ ; [1 mark]

- In this case, you need to know the number of moles of  $\text{CO}_2$  to be able to calculate the volume of  $\text{CO}_2$

2d

To calculate the percentage yield:

- Moles  $\text{K}_2\text{CO}_3 = \frac{\text{Moles KO}_2}{2} = \frac{0.0703235}{2} = 0.0352$  moles; [1 mark]
- Theoretical mass of  $\text{K}_2\text{CO}_3 = \text{mols} \times \text{molar mass} = 0.0352 \times 138.2 = 4.86 \text{ g} / 4.8593\dots \text{g}$ ; [1 mark]
- Percentage yield =  $\frac{\text{actual mass}}{\text{theoretical mass}} \times 100 = \frac{2.61}{4.86} \times 100 = 53.71 \%$ ; [1 mark]

[Total: 3 marks]

- The percentage yield shows how much of a particular product you get from the reactants compared to the maximum theoretical amount that you can get:
  - **Remember:** Percentage yield =  $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100$
  - The actual yield is the number of moles or mass of product obtained experimentally and is given in the question
  - The theoretical yield is the number of moles or mass obtained by a reacting mass calculation
    - You need to calculate this using the information from part (c)

3a

The number of moles of caffeine in a can of coke is:

- Moles of caffeine =  $\frac{0.034}{194.19} = 1.75 \times 10^{-4}$  ; [1 mark]

[Total: 1 mark]

- Watch out for unit conversions in questions
- First, convert the mass into grams
  - $34 \text{ mg} \div 1000 = 0.034 \text{ g}$
- Then find the number of moles by dividing by  $M_r$

- The answer should be given to two significant figures, but you won't be penalised here as it is not asked for in the question, and a range is usually accepted

3b

To calculate the empirical formula of caffeine:

- Moles of carbon: 4.12  
**AND**  
Moles of hydrogen = 5.15  
**AND**  
Moles of nitrogen = 2.06; [1 mark]
- Percentage of oxygen = 16.47 %  
**AND**  
Moles of oxygen = 1.03; [1 mark]
- Ratio of C : H : N : O = 4 : 5 : 2 : 1  
**AND**  
Empirical formula =  $\text{C}_4\text{H}_5\text{N}_2\text{O}$ ; [1 mark]

**[Total: 3 marks]**

- The first step of this calculation is to calculate the missing percentage of oxygen
  - Percentage of oxygen =  $100 - 49.48 - 5.20 - 28.85 = 16.47\%$
- You then follow the standard steps of an empirical formula calculation:
- 

| Element                          | C                           | H                          | N                           | O                           |
|----------------------------------|-----------------------------|----------------------------|-----------------------------|-----------------------------|
| Value (%)                        | 49.48                       | 5.20                       | 28.85                       | 16.47                       |
| $A_r$                            | 12.0                        | 1.0                        | 14.0                        | 16.0                        |
| Moles = value / $A_r$            | $\frac{49.48}{12.0} = 4.12$ | $\frac{5.20}{1.0} = 5.20$  | $\frac{28.85}{14.0} = 2.06$ | $\frac{16.47}{16.0} = 1.03$ |
| Ratio (divide by smallest moles) | $\frac{4.12}{1.03} = 4$     | $\frac{5.20}{1.03} = 5.05$ | $\frac{2.06}{1.03} = 2$     | $\frac{1.03}{1.03} = 1$     |

- This gives the final empirical formula of  $\text{C}_4\text{H}_5\text{N}_2\text{O}$
- In questions like this, you may get full marks for just writing the answer
  - However, if you make a mistake you will get nothing
  - This is why showing your working is so important as it can allow you to gain some marks even if you make an error



3c

The molecular formula of caffeine is:

- $C_8H_{10}N_4O_2$ ; [1 mark]

[Total: 1 mark]

- Divide the empirical mass by the molar mass to get the multiple
  - Empirical mass =  $(4 \times 12.01) + (5 \times 1.01) + (2 \times 14.01) + (1 \times 16.00) = 97.11$
  - Multiple =  $\frac{194.19}{97.11} = 1.99 \sim 2$
- Therefore, the molecular formula =  $2 \times$  empirical formula =  $2 \times C_4H_5N_2O = C_8H_{10}N_4O_2$

4a

To calculate the maximum mass of sodium carbonate that can be obtained from 0.75 kg of sodium chloride:

- $M_r$  of NaCl = 58.5  
**AND**  
Moles of NaCl = 12.821; [1 mark]
- Moles of NaHCO<sub>3</sub> = 12.821  
**AND**  
Moles of Na<sub>2</sub>CO<sub>3</sub> = 6.410; [1 mark]
- $M_r$  of Na<sub>2</sub>CO<sub>3</sub> = 106.0  
**AND**  
Mass of Na<sub>2</sub>CO<sub>3</sub> = 679 (g); [1 mark]

[Total: 3 marks]

- For this calculation, you will need the  $M_r$  of sodium chloride and sodium carbonate
  - NaCl =  $23.0 + 35.5 = 58.5$
  - Na<sub>2</sub>CO<sub>3</sub> =  $(23.0 \times 2) + 12.0 + (16.0 \times 3) = 106.0$
- Moles of sodium chloride =  $\frac{750}{58.5} = 12.821$
- From the first equation, one mole of sodium chloride produces one mole of sodium hydrogen carbonate
  - Therefore, 12.821 moles of sodium chloride will produce 12.821 moles of

## sodium carbonate

- From the second equation, two moles of sodium hydrogen carbonate produce one mole of sodium carbonate
  - Therefore, 12.821 moles of sodium hydrogen carbonate will produce  $\frac{12.821}{2} = 6.410$  moles of sodium carbonate
- Finally, mass of sodium carbonate = moles  $\times M_r = 6.410 \times 106.0 = 679$  g

4b

To calculate the minimum mass of  $\text{CaCO}_3$  that should be added to react with  $37.4 \text{ cm}^3$  of  $0.531 \text{ mol dm}^{-3}$  nitric acid:

- Equation =  $\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$ ; [1 mark]
- Moles of nitric acid = 0.01986; [1 mark]
- Moles of calcium carbonate = 0.00993

**AND**

$M_r$  of calcium carbonate =  $40.1 + 12.0 + (16.0 \times 3) = 100.1$ ; [1 mark]

- Mass of calcium carbonate = 0.994 (g)

**AND**

Answer given to 3 significant figures; [1 mark]

**[Total: 4 marks]**

- You must have a balanced chemical equation to work with for this calculation
- Remember:** Carbonate + acid  $\rightarrow$  salt + carbon dioxide + water
  - $\text{CaCO}_3 + 2\text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{CO}_2 + \text{H}_2\text{O}$
- Using the information in the question, you calculate the moles of nitric acid
  - Moles = concentration  $\times$  volume (in  $\text{dm}^3$ ) =  $0.531 \times \frac{37.4}{1000} = 0.01986$
- From your balanced chemical equation, two moles of nitric acid reacts with one mole of calcium carbonate
- Therefore, the moles of calcium carbonate =  $\frac{0.01986}{2} = 0.00993$
- Mass of calcium carbonate = moles  $\times M_r$ 
  - $M_r$  of calcium carbonate =  $40.1 + 12.0 + (16.0 \times 3) = 100.1$
  - Mass =  $0.00993 \times 100.1 = 0.994$  g
- Remember:** The question asks for an answer to an appropriate number of significant figures

- Since the values in the question are given to 3 significant figures, your answer should be to 3 significant figures

4c

To calculate the mass of anhydrous zinc chloride that could be obtained from the products of this reaction, assuming a 60.6% yield:

- Equation =  $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$ ; [1 mark]
- Moles of HCl = 0.0675

**AND**

Moles of zinc chloride = 0.03375; [1 mark]

- $M_r$  of zinc chloride = 136.4

**AND**

Mass of zinc chloride = 4.6035 (g); [1 mark]

- Mass of zinc chloride with 60.6% yield = 2.8 (g); [1 mark]

**[Total: 4 marks]**

- Examiners often comment about how students struggle with percentages that are incorporated into calculations
- You must have a balanced chemical equation to work with for this calculation
- **Remember:** Metal oxide + acid  $\rightarrow$  salt + water
  - $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$
- Using the information in the question, you calculate the moles of hydrochloric acid
  - Moles = concentration  $\times$  volume (in  $\text{dm}^3$ ) =  $0.90 \times \frac{75.0}{1000} = 0.0675$
- From your balanced chemical equation, two moles of hydrochloric acid produce one mole of zinc chloride
  - Therefore, the moles of zinc chloride =  $\frac{0.0675}{2} = 0.03375$

should be to 3 significant figures

4c

To calculate the mass of anhydrous zinc chloride that could be obtained from the products of this reaction, assuming a 60.6% yield:

- Equation =  $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$ ; [1 mark]

- Moles of HCl = 0.0675

**AND**

Moles of zinc chloride = 0.03375; [1 mark]

- $M_r$  of zinc chloride = 136.4

**AND**

Mass of zinc chloride = 4.6035 (g); [1 mark]

- Mass of zinc chloride with 60.6% yield = 2.8 (g); [1 mark]

**[Total: 4 marks]**

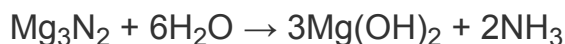
- Examiners often comment about how students struggle with percentages that are incorporated into calculations
- You must have a balanced chemical equation to work with for this calculation
- Remember:** Metal oxide + acid  $\rightarrow$  salt + water
  - $\text{ZnO} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{O}$
- Using the information in the question, you calculate the moles of hydrochloric acid
  - Moles = concentration  $\times$  volume (in  $\text{dm}^3$ ) =  $0.90 \times \frac{75.0}{1000} = 0.0675$
- From your balanced chemical equation, two moles of hydrochloric acid produce one mole of zinc chloride
  - Therefore, the moles of zinc chloride =  $\frac{0.0675}{2} = 0.03375$
- Mass of zinc chloride = moles  $\times M_r$ 
  - $M_r$  of zinc chloride =  $65.4 + (35.5 \times 2) = 136.4$
  - Mass =  $0.03375 \times 136.4 = 4.6035 \text{ g}$
- You then have to factor the 60.6% yield into your answer
  - $\frac{4.6035}{100} \times 60.6 = 2.8 \text{ g}$
- Remember:** The question does not ask for a specific number of significant figures
  - But, you should be giving all calculation answers to an appropriate number of significant figures

- Examiners will not always specify the number of significant figures but an appropriate number is expected

4d

To calculate the number of molecules of ammonia gas produced by the reaction of 12.62 g of magnesium nitride:

Balanced chemical equation:



- Correct chemical formulae; [1 mark]
- Correct balancing; [1 mark]

Calculation:

- $M_r$  of  $\text{Mg}_3\text{N}_2 = 100.9$   
**AND**  
Moles of  $\text{Mg}_3\text{N}_2 = 0.125$ ; [1 mark]
- Moles of  $\text{NH}_3 = 0.250$ ; [1 mark]
- Number of molecules of  $\text{NH}_3 = 1.51 \times 10^{23}$ ; [1 mark]

[Total: 5 marks]

- As with the previous parts of this question, you must have a balanced chemical equation to work with for this calculation
- The question tells you that magnesium nitride,  $\text{Mg}_3\text{N}_2$ , reacts with water to form magnesium hydroxide and ammonia
  - $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{NH}_3$
- The three magnesium atoms in magnesium nitride will form  $3\text{Mg}(\text{OH})_2$
- The two nitrogen atoms in magnesium nitride will form  $2\text{NH}_3$ 
  - $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \rightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
- There are 6 hydroxide ions in the  $3\text{Mg}(\text{OH})_2$  and 6 hydrogen atoms in the  $2\text{NH}_3$ , which require  $6\text{H}_2\text{O}$ 
  - $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
- Using the information in the question, you calculate the moles of magnesium nitride
  - $M_r$  of  $\text{Mg}_3\text{N}_2 = (24.3 \times 3) + (14.0 \times 2) = 100.9$

- $\text{Moles} = \frac{\text{mass}}{M_r} = \frac{12.62}{100.9} = 0.125$
- From your balanced chemical equation, two moles of ammonia are produced from one mole of magnesium nitride
  - Therefore, the moles of ammonia =  $0.125 \times 2 = 0.25$
- One mole of any chemical contains  $6.02 \times 10^{23}$  particles / molecules
  - Therefore, 0.25 moles contains  $0.25 \times 6.02 \times 10^{23} = 1.51 \times 10^{23}$  particles / molecules

5a

i) The formula of calcium phosphate is:

- $\text{Ca}_3(\text{PO}_4)_2$ ; [1 mark]

ii) The oxidation number of calcium in hydroxyapatite is:

- +2; [1 mark]
- (Because) the phosphate ion is  $-3$

**AND**

The hydroxyl / hydroxide ion is  $-1$ ; [1 mark]

- The overall charge of the compound / hydroxyapatite is 0

**OR**

$\text{Ca}_5(\text{PO}_4)_3\text{OH} = (+2 \times 5) + (-3 \times 3) + (-1) = 0$ ; [1 mark]

iii) The formula of chlorapatite is:

- $\text{Ca}_5(\text{PO}_4)_3\text{Cl}$ ; [1 mark]

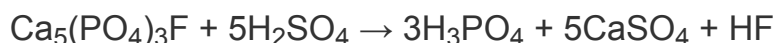
**[Total: 5 marks]**

- In part (i)
  - Calcium is in Group 2 so it is a  $2+$  ion,  $\text{Ca}^{2+}$
  - You should know that the phosphate ion is a  $3-$  ion,  $\text{PO}_4^{3-}$
  - This means that you need three  $\text{Ca}^{2+}$  ions and two  $\text{PO}_4^{3-}$  ions to get the electronically neutral calcium phosphate
- In part (ii)
  - It is easy to say that the oxidation number of calcium is  $+2$
  - But, this will not gain full marks because you are asked to explain the answer

- You need to talk through the charges of the phosphate and hydroxide ions and show how they balance out the charge of 5 calcium ions
- In part (iii)
  - You are given the formula of hydroxyapatite, so you can replace the hydroxide ion with a chloride ion to suggest a formula for chlorapatite

5b

The balanced chemical equation for the reaction of fluorapatite,  $\text{Ca}_5(\text{PO}_4)_3\text{F}$ , with sulfuric acid to form hydrogen phosphate, calcium sulfate and hydrogen fluoride is:



- Correct chemical formulae; [1 mark]
- Correct balancing; [1 mark]

**[Total: 2 marks]**

- You should know or be able to deduce the formulae of the products
  - Hydrogen phosphate contains  $\text{H}^+$  ions and  $\text{PO}_4^{3-}$  ions
    - So, you need 3  $\text{H}^+$  ions to balance out one  $\text{PO}_4^{3-}$  ion
  - Calcium sulfate contains  $\text{Ca}^{2+}$  ions and  $\text{SO}_4^{2-}$  ions
    - So, you need one of each ion
  - Hydrogen fluoride contains  $\text{H}^+$  ions and  $\text{F}^-$  ions
    - So, you need one of each ion
- This gives an unbalanced equation as a starting point
  - $\text{Ca}_5(\text{PO}_4)_3\text{F} + \text{H}_2\text{SO}_4 \rightarrow \text{H}_3\text{PO}_4 + \text{CaSO}_4 + \text{HF}$
- $\text{Ca}_5(\text{PO}_4)_3\text{F}$  contains 5 calcium and 3 phosphates, which suggests that 5 calcium sulfate and 3 hydrogen phosphate could be formed
  - $\text{Ca}_5(\text{PO}_4)_3\text{F} + \text{H}_2\text{SO}_4 \rightarrow 3\text{H}_3\text{PO}_4 + 5\text{CaSO}_4 + \text{HF}$
- The production of 5 calcium sulfates would require 5 sulfuric acids
  - $\text{Ca}_5(\text{PO}_4)_3\text{F} + 5\text{H}_2\text{SO}_4 \rightarrow 3\text{H}_3\text{PO}_4 + 5\text{CaSO}_4 + \text{HF}$
- This completes the balanced equation
  - Although it is always worth double-checking your answer with unfamiliar and more complicated chemical equations

5c

To calculate the mass of calcium dihydrogenphosphate formed when 80.2 g of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  reacts with 87.9 g  $\text{H}_2\text{SO}_4$ :

Moles calculations:

- $M_r$  of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  = 504.5  
**AND**  
Moles of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  = 0.0995; [1 mark]
- $M_r$  of  $\text{H}_2\text{SO}_4$  = 98.1  
**AND**  
Moles of  $\text{H}_2\text{SO}_4$  = 0.89602; [1 mark]

Limiting reagent calculation:

- 0.0995 moles of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  reacts with 0.69653 moles of  $\text{H}_2\text{SO}_4$  **AND**  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  is the limiting reagent  
**OR**  
0.89602 moles of  $\text{H}_2\text{SO}_4$  reacts with 0.12800 moles of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  **AND**  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  is the limiting reagent; [1 mark]

$\text{Ca}(\text{H}_2\text{PO}_4)_2$  calculations:

- 0.0995 moles of  $\text{Ca}_5(\text{PO}_4)_3\text{F}$  makes 0.14926 moles of  $\text{Ca}(\text{H}_2\text{PO}_4)_2$   
**AND**  
 $M_r$  of  $\text{Ca}(\text{H}_2\text{PO}_4)_2$  = 234.1; [1 mark]
- Mass of  $\text{Ca}(\text{H}_2\text{PO}_4)_2$  = 34.9 (g); [1 mark]

**[Total: 5 marks]**

- This question is a reacting masses question using an unfamiliar equation, limiting reagent calculations and a variety of stoichiometry
- Initially, you need to calculate the  $M_r$  and number of moles of each reactant to